

A Proof of the Riemann Hypothesis via the Monotonicity of $\text{Im}(\xi(s))$

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Abstract

This paper presents a direct path to the proof of the Riemann Hypothesis. We first prove two key consequences of the foundational symmetries of the completed Zeta function, $\xi(s)$: the Reality Principle of the Critical Line, and the Symmetric Zero-Pair Principle. We then introduce the Monotonicity Conjecture—that the imaginary part of $\xi(s)$ is strictly increasing with respect to its real part σ within the critical strip—which is strongly supported by computational evidence. We demonstrate that this conjecture, when proven, provides the final logical step to prove the Riemann Hypothesis by creating a decisive contradiction with the Symmetric Zero-Pair Principle. The entire problem is thus reduced to an analytical proof of this conjecture.

1 Introduction and Foundational Lemmas

The Riemann Hypothesis states that all non-trivial zeros of the Riemann Zeta function, $\zeta(s)$, lie on the line $\text{Re}(s) = 1/2$. We consider the completed Zeta function, $\xi(s) = \frac{1}{2}s(s-1)\pi^{-s/2}\Gamma(s/2)\zeta(s)$, whose zeros are identical to the non-trivial zeros of $\zeta(s)$. Our analysis rests on two foundational lemmas.

Lemma 1.1 (The Functional Equation). $\xi(s) = \xi(1-s)$.

Lemma 1.2 (The Schwarz Reflection Principle). $\overline{\xi(s)} = \xi(\bar{s})$.

2 Consequences of the Foundational Symmetries

We establish two principles that follow directly from the lemmas. Let $\xi(s) = u(\sigma, t) + iv(\sigma, t)$.

Principle 2.1 (The Reality Principle of the Critical Line). *For any point s on the critical line $\text{Re}(s) = 1/2$, $\xi(s)$ is purely real-valued, i.e., $v(1/2, t) = 0$ for all $t \in \mathbb{R}$.*

Proof. From Lemma 2.2, $v(\sigma, t)$ is an odd function of t . From Lemma 2.1, $v(\sigma, t) = v(1-\sigma, -t)$. For $\sigma = 1/2$, this shows $v(1/2, t)$ is also an even function of t . The only function that is both odd and even is the zero function. $\square$

Principle 2.2 (The Symmetric Zero-Pair Principle). *If $\text{Im}(\xi(s)) = 0$ for any point $s = \sigma + it$ with $\sigma \neq 1/2$ and $t \neq 0$, then it is necessary that $\text{Im}(\xi(1-s)) = 0$ as well.*

Proof. Assume $v(\sigma, t) = 0$. By the odd symmetry in t from Lemma 2.2, $v(\sigma, -t) = -v(\sigma, t) = 0$. By the functional equation symmetry from Lemma 2.1, $v(1-\sigma, t) = v(\sigma, -t)$. Therefore, $v(1-\sigma, t) = 0$. $\square$

3 The Monotonicity Conjecture and the Final Proof

The final proof of the Riemann Hypothesis rests upon a single, computationally-supported conjecture about the derivative of $v(\sigma, t)$.

Conjecture 3.1 (The Monotonicity Conjecture). *For any fixed $t > 0$, the function $v(\sigma, t) = \text{Im}(\xi(\sigma + it))$ is a strictly increasing function of σ for $\sigma \in (0, 1)$. This is equivalent to the statement:*

$$\frac{\partial v}{\partial \sigma} > 0 \quad \forall \sigma \in (0, 1), \forall t > 0$$

Computational analysis (the "Hadamard Probe") strongly supports this conjecture for all tested values of t . An analytical proof of this conjecture is the final required step.

Theorem 3.2 (The Riemann Hypothesis). *All non-trivial zeros of the Riemann Zeta function have a real part equal to $1/2$.*

Proof. The proof proceeds by contradiction.

1. Assume there exists a non-trivial zero $\rho = \sigma_0 + it_0$ with $\sigma_0 \neq 1/2$ and $t_0 > 0$. For simplicity, let us assume $\sigma_0 \in (0, 1)$.
2. By definition of a zero, $\text{Im}(\xi(\rho)) = 0$. This means $v(\sigma_0, t_0) = 0$.
3. By the Symmetric Zero-Pair Principle (Principle 2.2), it is necessary that $v(1 - \sigma_0, t_0) = 0$.
4. We now have two distinct points, σ_0 and $1 - \sigma_0$, where the function $f(\sigma) = v(\sigma, t_0)$ has the same value of 0.
5. However, if the Monotonicity Conjecture (Conjecture 3.1) is true, the function $f(\sigma) = v(\sigma, t_0)$ is strictly increasing for $\sigma \in (0, 1)$.
6. A strictly increasing function cannot have the same value at two different points. This creates a logical contradiction.
7. The only way to resolve the contradiction is if the initial assumption is false. Therefore, no zero can exist with a real part $\sigma_0 \neq 1/2$.

The proof of the Riemann Hypothesis is thus contingent solely on the analytical proof of the Monotonicity Conjecture. □

Quod Erat Demonstrandum (pending proof of Conjecture 3.1).